

Section 12 Solution

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3 Exercise

3.1 Production with Fixed Costs

Suppose a firm has a cost function $c(y) = H + y^2$ if $y > 0$ and $c(0) = 0$.

1. Derive and draw the MC and AC curves.

Answer: $MC = C'(y) = 2y$, $AC = C(y)/y = H/y + y$. See graph below on the left.

2. Solve for y at the point where the two curves intersect, and show that this point is also the minimum of the AC curve.

Answer: Set $AC = MC$, you get $y^* = \sqrt{H}$. You can use FOC and SOC to check that it is the lowest point of AC .

In fact, MC and AC always intersect at AC 's minimum (if they intersect). Think about intuition (discussed in section). We can prove it using the quotient rule to differentiate $\frac{c(y)}{y}$ with respect to y (this is the f.o.c. for finding the minimum of $\frac{c(y)}{y}$):

$$\frac{\partial}{\partial y} \left(\frac{C(y)}{y} \right) = \frac{C'(y_{min})y_{min} - C(y_{min})}{y_{min}^2} = 0$$

$$C'(y_{min}) = \frac{C(y_{min})}{y_{min}}$$

3. Draw the supply function for the firm with cost function $c(y) = H + y^2$ if $y > 0$ and $c(0) = 0$, and write an expression representing it: $(y^*(p))$.

Answer:

See graph above on the right.

$$y^*(p) = \begin{cases} p/2 & \text{if } p \geq 2\sqrt{H} \\ 0 & \text{if } p < 2\sqrt{H} \end{cases}$$

4. If p stays constant, how does y^* respond to an increase in H ?

Answer:

Say that H increases to H' . There are three possible scenarios:

- $2\sqrt{H'} \leq p$, then y^* stays at $p/2$.
- $2\sqrt{H} \leq p < 2\sqrt{H'}$, then y^* falls from $p/2$ to 0.
- $p < 2\sqrt{H}$, then y^* stays at 0.

5. What is the profit function of this firm?

Answer:

$$\pi^*(p) = \begin{cases} py - H - y^2 & \text{if } p \geq 2\sqrt{H} \\ 0 & \text{if } p < 2\sqrt{H} \end{cases} = \begin{cases} p^2/4 - H & \text{if } p \geq 2\sqrt{H} \\ 0 & \text{if } p < 2\sqrt{H} \end{cases}$$

*These notes are a consolidation of those of Jianmeng Lyu, Sam Leone, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Nick Li, Katalin Springel, Jonas Tungodden, Dan Acland, Justin Gallagher, Mariana Carrera, Matt Leister, Ivan Balbuzanov, Anne Karing, and David Wu, Sam Wang, and Matteo Saccarola.

3.2 Market Equilibrium

Consider an economy of 3 individuals and 2 firms producing one good.

- Individual 1 has demand function: $x_1 = 500 - p$
- Individual 2 has demand function: $x_2 = 750 - p$
- Individual 3 has demand function: $x_3 = 250 - p$
- Firm 1 has supply function: $Y_1 = 300 + 4p$
- Firm 2 has supply function: $Y_2 = 700 + 2p$

Answer the following questions:

1. What is the aggregate demand? **Answer:**

$$X^D(p) = x_1(p) + x_2(p) + x_3(p) = \begin{cases} 0 & \text{if } p > 750 \\ 750 - p & \text{if } p \in (500, 750] \\ 1250 - 2p & \text{if } p \in (250, 500] \\ 1500 - 3p & \text{if } p \in [0, 250] \end{cases}$$

2. What is the aggregate supply? **Answer:** $Y^S(p) = Y_1(p) + Y_2(p) = 1,000 + 6p$
3. What is the equilibrium price?

Answer: We are not sure in which section of the piecewise aggregate demand function the aggregate supply intersect. To check this we try with each segment, until we find one equilibrium price that belongs to the segment:

- $750 - p = 1,000 + 6p$ and then $p^* = -250/7 = -35.7 \rightarrow$ does not belong to $(500, 750]$
- $1250 - 2p = 1,000 + 6p$ and then $p^* = 250/8 = 31.25 \rightarrow$ does not belong to $(250, 500]$
- $1,500 - 3p = 1,000 + 6p$ and then $p^* = 500/9 = 55.56 \rightarrow$ does belong to $[0, 250]$. The aggregate supply intersects the aggregate demand in this section!

4. What is the equilibrium quantity sold in the market?

Answer: $y^* = 1000 + 6p^* = 1,333.3$

Demand, Supply, and Taxes (Problem 1, MT2, Spring 2012)

Assume the case of perfect competition, with each firm i producing quantity q_i with total cost $C_i(q_i) = cq_i$. That is, costs are linear in the quantity produced.

1. Determine the marginal cost function C'_q and the average cost function $C(q)/q$, and plot the two functions in a graph with x-axis quantity q_i and y-axis cost/price.

Answer:

The marginal cost and average cost both equal c : $C'_q = C(q)/q = c$.

2. Plot graphically the supply function for each firm. Also, write it analytically, in the form $q^* = S(p)$, that is, what is the quantity supplied q^* as a function of p ?

Answer:

The supply function is

$$S(p) = \begin{cases} +\infty & \text{if } p > c \\ \text{any quantity } q \in [0, \infty) & \text{if } p = c \\ 0 & \text{if } p < c \end{cases}$$

3. Assume now that in perfect competition there are 5 firms, all with the same cost function $C_i(q_i) = cq_i$. How does the aggregate supply function differ from the individual supply function, if at all? Plot, and write analytically.

Answer:

The aggregate supply function which comes from horizontal aggregation in this case is exactly identical.

4. Assume now that aggregate demand is given by the linear (inverse) demand function $p(Q) = A - bQ$, with $A > c$. Draw in the graph with the marginal cost function of point (3). If you want, assume $A = 10$, $c = 5$, $b = 1$. Solve (graphically or otherwise) for the equilibrium perfect competition price p_{PC}^* , as well as for the overall quantity produced Q^* , and obtain also analytical solutions. Notice that the equilibrium will be where demand equates supply. How do quantity and price depend on A and c ?

Answer:

Demand will equate supply for a price p_{PC}^* equal to c . Given this price, the quantity produced will be given by the demand function, and hence

$$\begin{aligned} p_{PC}^* &= c = A - bQ, \text{ and hence} \\ Q_{PC}^* &= \frac{A - c}{b} \end{aligned}$$

The equilibrium price is increasing in marginal cost c , but independent of the demand intercept A . The equilibrium quantity is increasing in the demand intercept A and decreasing in marginal cost c .

